

Measures Of Central Tendency
OR
Measures of Location
OR
Average

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Measures of Central Tendency

- The diagrammatic representation of a set of data can give us some impression about its distribution.
- Even then there remains a need for a single quantitative measure which could be used to indicate the center of the distribution
- Tendency of the values to concentrate at their center is central tendency and any measure indicating the center of their distribution is called a measure of central tendency.
- An average is a single value that summarizes the data.

Types of Measures of Central Tendency

The most common measures of central tendency are:

- 1. Arithmetic Mean (AM)/ Mean**
- 2. Geometric Mean (GM)**
- 3. Harmonic Mean (HM)**
- 4. Median & Quantiles**
- 5. Mode**

The decision as to which measure is to be used depends on the particular situation under consideration.

Arithmetic Mean

The arithmetic mean is defined as a value obtained by dividing the sum of all the observations by their number

$$\text{Arithmetic Mean} = \frac{\text{Sum of all the observations}}{\text{Number of the observations}}$$

If $X_1, X_2, \dots, X_n$ are n observations of a variable X then their AM is defined as:

$$\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n} = \frac{\sum X}{n} \quad n\bar{X} = \sum X$$

Arithmetic Mean

The marks obtained by 8 students are:

67	72	68	70	65	68	75	63
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Find MEAN. $\bar{X} = \frac{67+72+\dots+63}{8} = \frac{\sum X}{n} = \frac{548}{8} = 68.5$ Marks

If we assume each value equal to mean of the data (68.5)

68.5	68.5	68.5	68.5	68.5	68.5	68.5	68.5
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$$8 \times 68.5 = 548$$

$$n\bar{X} = \sum X$$

Properties of AM

1-The sum of deviations of observations from mean is always equal to zero

$$\sum (X - \bar{X}) = 0$$

2-The sum of squared deviations of observations from mean is minimum

$$\sum (X - \bar{X})^2 < \sum (X - a)^2$$

Where 'a' is any value other than mean of the data

3-If a variable X has mean $\bar{X}$ then mean of new variable Y will be

$$\text{If } Y = a + bX$$

$$\bar{Y} = a + b\bar{X}$$

Properties of AM

X	(X-68.5)	(X-68.5) ²	(X-70)	(X-70) ²	Y=2X+3
67	-1.5	2.25	-3	9	137
72	3.5	12.25	2	4	147
68	-0.5	0.25	-2	4	139
70	1.5	2.25	0	0	143
65	-3.5	12.25	-5	25	133
68	-0.5	0.25	-2	4	139
75	6.5	42.25	5	25	153
63	-5.5	30.25	-7	49	129
	0	102	-12	120	1120

Mean of Y=1120/8=140

Mean of Y=2(68.5)+3=140

Problems related to AM

EXAMPLE:-The mean weight of 10 students is 50 Kg when two students left the class the mean weight becomes 48 Kg find the mean weight of students who left the class

SOLUTION:-

Total weight of 10 student $= (10)(50) = 500$ Kg

Total weight of 8 student (after 2 students left the class)
 $= (8)(48) = 384$ Kg

Total weight of 2 students (Who left the class)
 $= 500 - 384 = 116$ Kg

Mean weight of the students who left the class
 $= 116 / 2 = 58$ kg

Problems related to AM

EXAMPLE:- For a class of 25 students, on Tuesday 20 students from the class took a Math test and their mean marks was 80. On Friday remaining students from the class took the Math test and their mean marks was 90.

Find the mean marks of the entire class ?

SOLUTION:-

Total marks of 20 students who took test on Tuesday = $(20)(80) = 1600$

Total marks of 5 students who took test on Friday = $(5)(90) = 450$

Total marks of 25 students = $1600 + 450 = 2050$

Mean marks of 25 students = $2050/25 = 82$

Classes	Freq (f)	Mid-Point (X)	Total height from each group (f)x(X)
86–90	6	88	6x88=528
91–95	4	93	372
96–100	10	98	980
101–105	6	103	618
106–110	3	108	324
111–115	1	113	113
Total	30		2935

Mean using actual data $\bar{X} = 97.63$ $\bar{X} = \frac{2935}{30} = 97.83$ 10

Weighted Arithmetic Mean

Sometimes, all values aren't of equal importance, and we associate with them certain numerical values as weighting factors to express their relative importance.

These numerical values are known as weights. We assign weights $W_1, W_2, \dots, W_k$ to the observations according to their relative importance.

$$\bar{X}_w = \frac{\sum WX}{\sum W}$$

EXAMPLE:- An examination was held to decide about the award of a scholarship in an institution. The weights of various subjects were different. The marks obtained by candidates A and B are given below. If the candidate getting the highest average score is to be awarded the scholarship, who should get it

Subject	Weights (%)	X_A	X_B	WX_A	WX_B
Statistics	40	70	85	2800	3400
Mathematics	30	90	75	2700	2250
Economics	20	50	45	1000	900
English	10	60	65	600	650
TOTAL	100	270	270	7100	7200

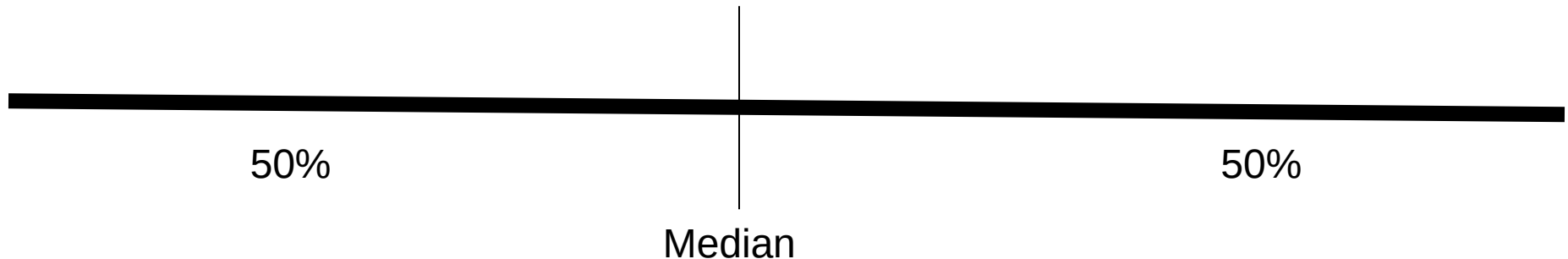
UNWEIGHTED MEAN(AM)
A= 67.5
B= 67.5

WEIGHTED MEAN
A= 71
B= 72

Median

Median is the value that divides the arranged data into 2 equal parts i.e 50% data below and 50% data above it i.e median is at center of the data

$$\text{Median} = \frac{(n+1)^{\text{th}}}{2} \text{ observation in the data}$$



Median

The following data represent the weekly TV viewing time in hours for 20 respondents, calculate TV viewing time of 50% (10) respondents is less than how many hours

5	15	16	20	21	25	26	27	30	30
31	32	32	34	35	37	38	41	43	66

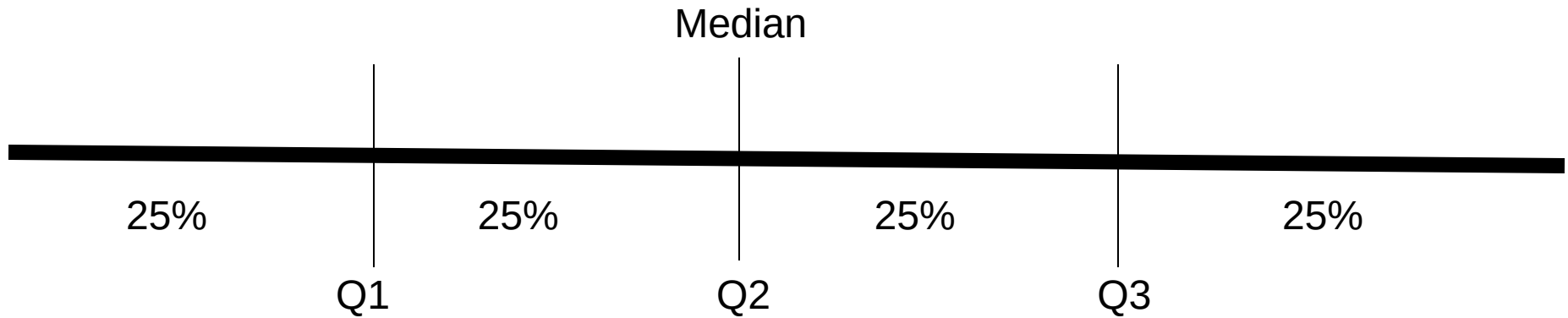
LOCATION $Median = Size\ of\ \frac{(n+1)^{th}}{2}$ observation = 10.5th observation

$$\begin{aligned} Median &= 10^{th} + 0.5(11^{th} - 10^{th}) \\ &= 30 + 0.5(31 - 30) = 30.5 \text{ hours} \end{aligned}$$

Quartiles

Quartiles (Q1, Q2, Q3) are the values that divide the arranged data into 4 equal parts

$$Q_j = \frac{j(n+1)^{\text{th}}}{4} \text{ observation in the data, } j=1, 2, 3$$



Quartiles

5	15	16	20	21	25	26	27	30	30
31	32	32	34	35	37	38	41	43	66

$$Q_1 = \frac{1(20+1)}{4}^{\text{th}} \text{ obs.} = 5.25^{\text{th}} \text{ obs.}$$

$$= 5^{\text{th}} + 0.25(6^{\text{th}} - 5^{\text{th}}) = 21 + 0.25(25 - 21) = 22.0 \text{ Hours}$$

$$Q_2 = \frac{2(20+1)}{4}^{\text{th}} \text{ obs.} = 10.50^{\text{th}} \text{ obs.}$$

$$= 10^{\text{th}} + 0.50(11^{\text{th}} - 10^{\text{th}}) = 30 + 0.50(31 - 30) = 30.5 \text{ Hours}$$

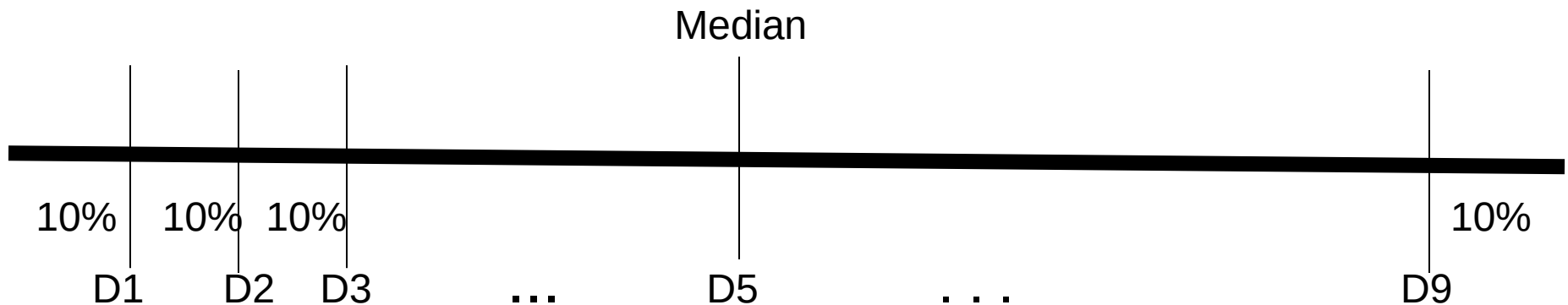
$$Q_3 = \frac{3(20+1)}{4}^{\text{th}} = 15.75^{\text{th}} \text{ obs.}$$

$$= 15^{\text{th}} + 0.75(16^{\text{th}} - 15^{\text{th}}) = 35 + 0.75(37 - 35) = 36.5 \text{ Hours}^{16}$$

Deciles

Deciles (D1, D2, D3, . . . D9) are the vales that divide the arranged data in to 10 equal parts

$$D_j = \frac{j(n+1)}{10}^{\text{th}} \text{ observation in the data, } j = 1, 2, \dots, 9$$



Deciles

5	15	16	20	21	25	26	27	30	30
31	32	32	34	35	37	38	41	43	66

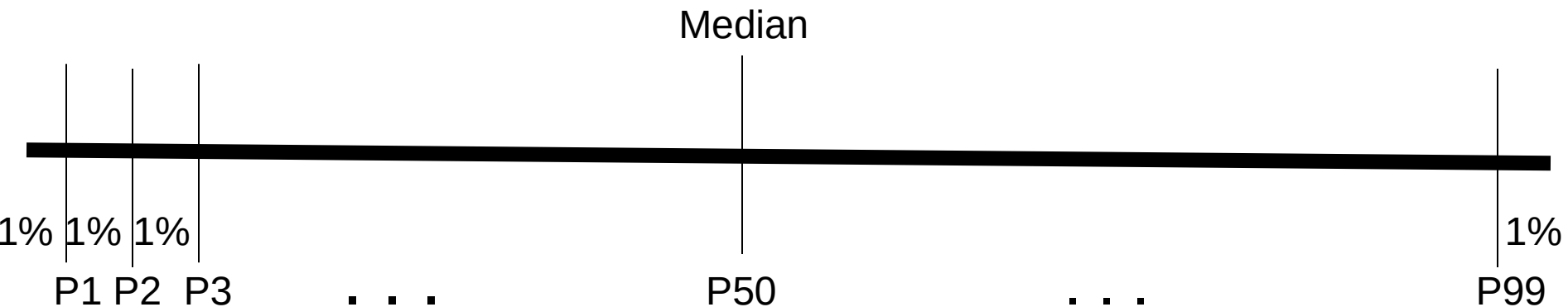
$$D4 = \frac{4(20+1)^{\text{th}}}{10} \text{ obs.} = 8.4^{\text{th}} \text{ obs.}$$

$$= 8^{\text{th}} + 0.4(9^{\text{th}} - 8^{\text{th}}) = 27 + 0.4(30 - 27) = 28.2 \text{ Hours}$$

Percentiles

Percentiles (P1, P2, P3, . . ., P99) are the values that divide the arranged data into 100 equal parts

$$P_j = \frac{j(n+1)}{100}^{\text{th}} \text{ observation in the data, } j = 1, 2, \dots, 99$$



P25=Q1

P50=Q2=Median

P75=Q3

P10=D1 , P20=D2, . . . P90=D9

Median and Quantiles for group data

The following data represent the cholesterol level in milligrams per 100 milliliters of 200 patients

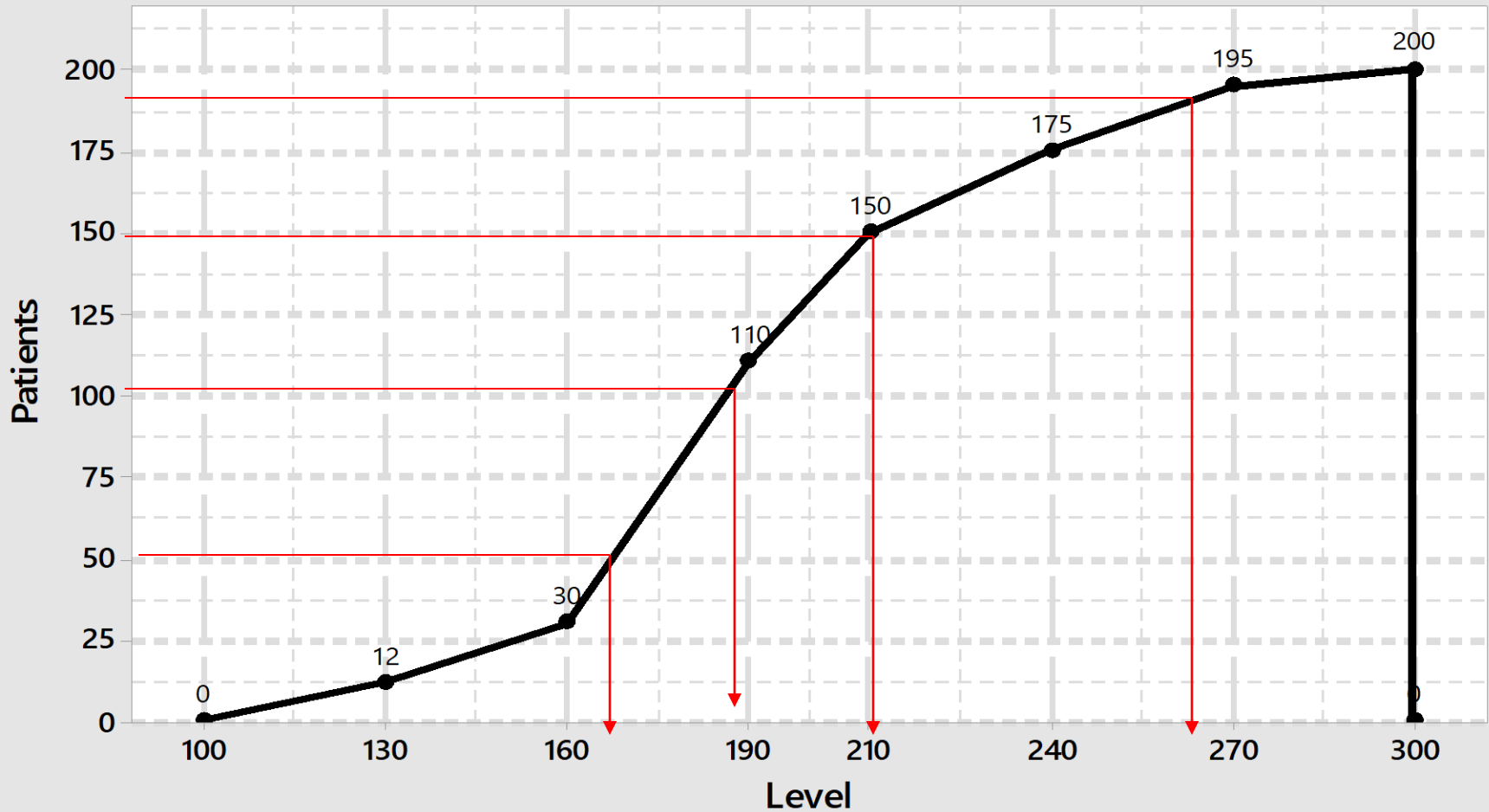
Level	f	c.f
100---130	12	12
130—160	18	30
160--190	80	110
190--210	40	150
210—240	25	175
240—270	20	195
270—300	05	200

If normal range of level is 150---200

- Can we conclude that 50% (100) patients level is lower than upper level of normal range
- Can we conclude that 25% (50) patients level is lower than lower level of normal range
- If level higher than 260 is at higher risk of heart attack Can we conclude that 25% (50) patients are at higher risk of heart attack
- If level higher than 280 is very much likely to suffer heart attack in near future. Can we conclude that 05% (10) patients are at very much likely to have heart attack in near future

- Level of bottom 50% (100) patients 188
- Level of bottom 25% (50) patients 165
- Level of top 25% (50) patients 210
- Level of top 05% (10) patients 265

Cumulative Frequency Polygon



Mode

A value which repeats most frequently in a set of data. That is, Mode is a value that repeats maximum number of times in a sequence of observations.

Mode

Listed below are the favorite colour of a sample of students. What is the mode?

W	G	W	B	W	G	R
W	R	G	R	G	G	B

Mode = Green

Because Green is repeated most (5) number of times

The following data represent the gender, marks out of 10 and height of 9 students in a class

Gender	Male	Female	Female	Male	Female	Male	Male	Male	Male
Marks	6	7	7	8	9	7	9	10	9
Height	5.5	5.7	4.9	5.0	5.3	4.8	5.6	4.7	5.8

Find mode of gender , marks and height

Solution:- Mode (gender)=Male , Mode(marks) = 7,9 Mode(Height)= No mode

NOTE:- (i) A data may have more than one mode or no-mode at all

(ii) A data with one mode is called uni-model, 2 modes bi-model or more than 2 modes multi model data

(iii) Mode is possible for both Qualitative and Quantitative data

Example:- The following data represent the purchase of different brand of software drink

Soft Drink	Frequency
Coke Classic	19
Diet Coke	8
7-up	5
Pepsi Cola	13
Sprite	5
TOTAL	50

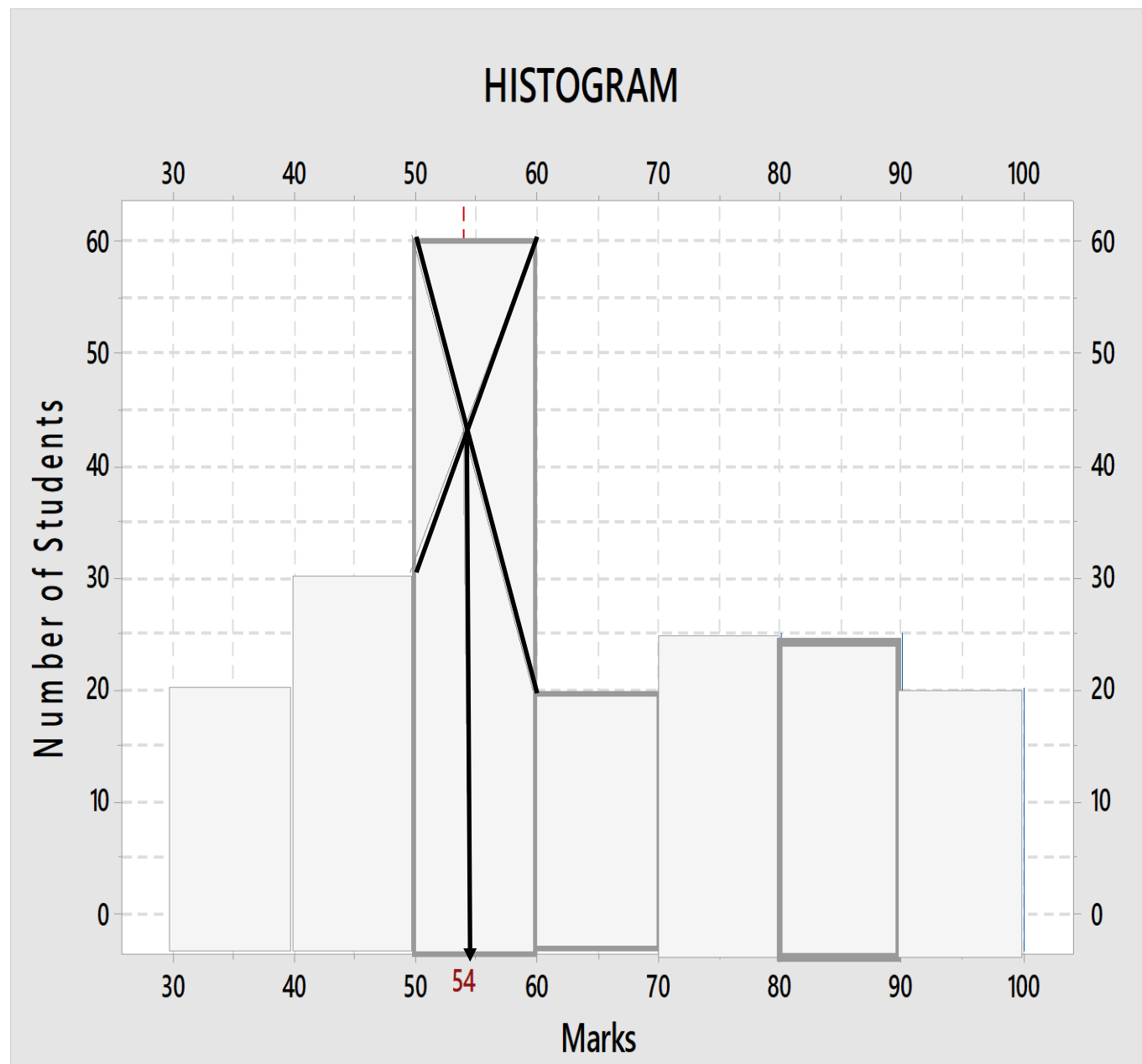
Mode = Coke Classic

i.e Coke Classic is the most frequently purchased soft drink

Mode for Group Data

The data represent the marks obtained by students, and Mode of the data graphically

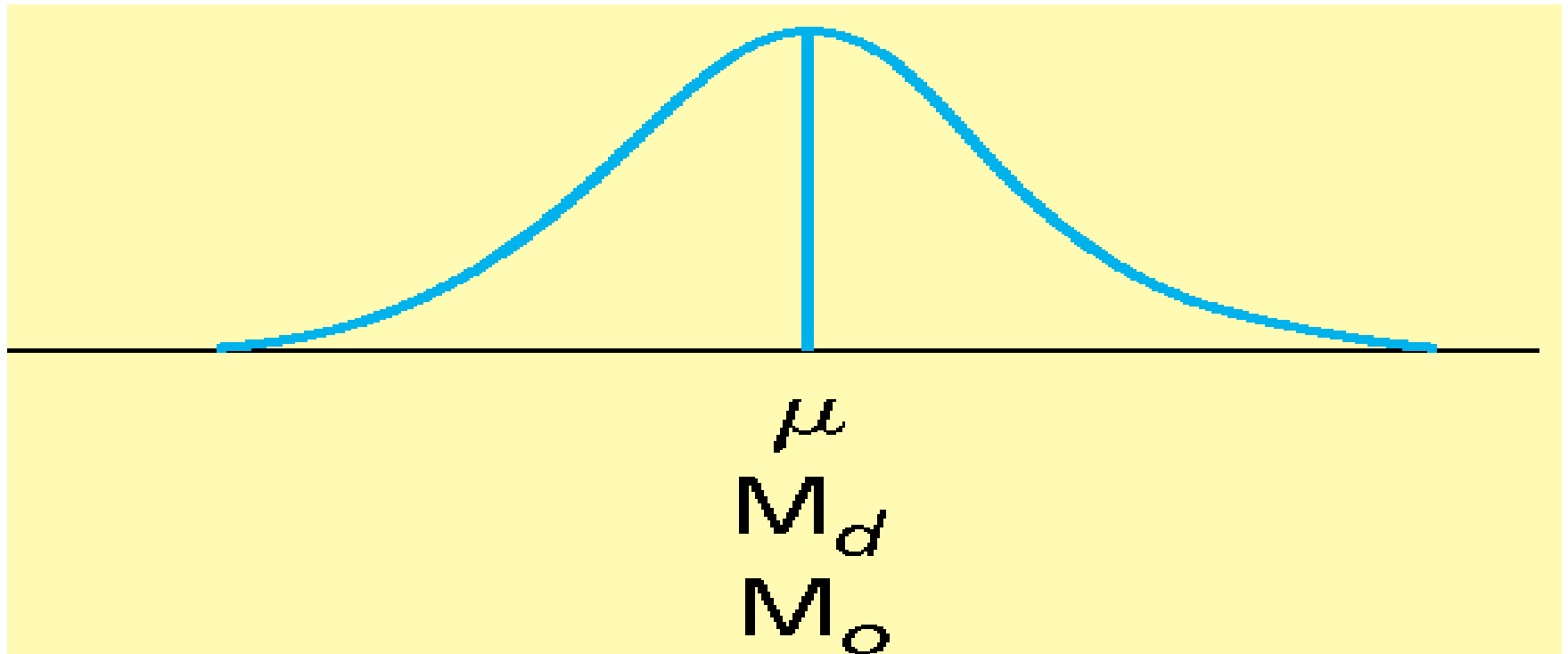
Marks	f
30----40	20
40----50	30
50----60	60
60----70	20
70----80	25
80----90	25
90--100	20



Relationship between Mean, Median and Mode

For a symmetrical data

$$\text{Mean} = \text{Median} = \text{Mode}$$



For Skewed Data

Positively skewed

$Mean > Median > Mode$



Negatively skewed

$Mode > Median > Mean$

